taken up by Martian microbes. To the initial astonishment of, I think, all the scientists involved, each of the three experiments gave what at first seemed to be positive results. Gases were exchanged; organic matter was oxidized; carbon dioxide was incorporated into the soil.

But there are reasons for caution. These provocative results are not generally thought to be good evidence for life on Mars: The putative metabolic processes of Martian microbes occurred under a very wide range of conditions inside the *Viking* landers—wet (with liquid water brought from Earth) and dry, light and dark, cold (only a little above freezing) to hot (almost the normal boiling point of water). Many microbiologists deem it unlikely that Martian microbes would be so capable under such varied conditions. Another strong inducement to skepticism is that a fourth experiment, to look for organic chemicals in the Martian soil, gave uniformly negative results despite its sensitivity. We expect life on Mars, like life on Earth, to be organized around carbon-based molecules. To find no such molecules at all was daunting for optimists among the exobiologists.

The apparently positive results of the life detection experiments is now generally attributed to chemicals that oxidize the soil, deriving ultimately from ultraviolet sunlight (as discussed in the previous chapter). There is still a handful of *Viking* scientists who wonder if there might be extremely tough and competent organisms very thinly spread over the Martian soil—so their organic chemistry could not be detected, but their metabolic processes could. Such scientists do not deny that ultraviolet-generated oxidants are present in the Martian soil, but stress that no thorough explanation of the *Viking* life detection results from oxidants alone has been forthcoming. Tentative claims have been made of organic matter in SNC meteorites, but they seem instead to be contaminants that have entered the meteorite after its arrival on our world. So far, there are no claims of Martian microbes in these rocks from the sky.

Perhaps because it seems to pander to public interest, NASA and most *Viking* scientists have been very chary about pursuing the biological hypothesis. Even now, much more could be done in going